

Topology

Server
Port: 36657
IP: 169.144.0.6

Client
Port: 39876
IP: 169.144.0.5

Message Table

M	Client1	Server	Client2	Hex
1	(SIP/SDP) Request: INVITE sip:2222@localhost.localdomain			Page 1
2		(SIP) Status: 407 Proxy Authentication Required		Page
3	(SIP) Request: ACK sip:2222@localhost.localdomain			Page
4	(SIP/SDP) Request: INVITE sip:2222@localhost.localdomain			Page
5		(SIP) Status: 100 Trying		Page
6		(SIP) Status: 180 Ringing		Page
7	(SIP/SDP) Status: 183 Session Progress			Page
8		(RTP) PT=ITU-T G.711 PCMU, SSRC=0x7D42AA5E, Seq=14659, Time=139096		
⋮				
9	(RTCP) Receiver Report Source description			Page
10		(SIP/SDP) Request: INVITE sip:2222@178.14.3.15:3269;ri nstance=8858914d0a515fa3		Page
11			(SIP) Status: 180 Ringing	Page

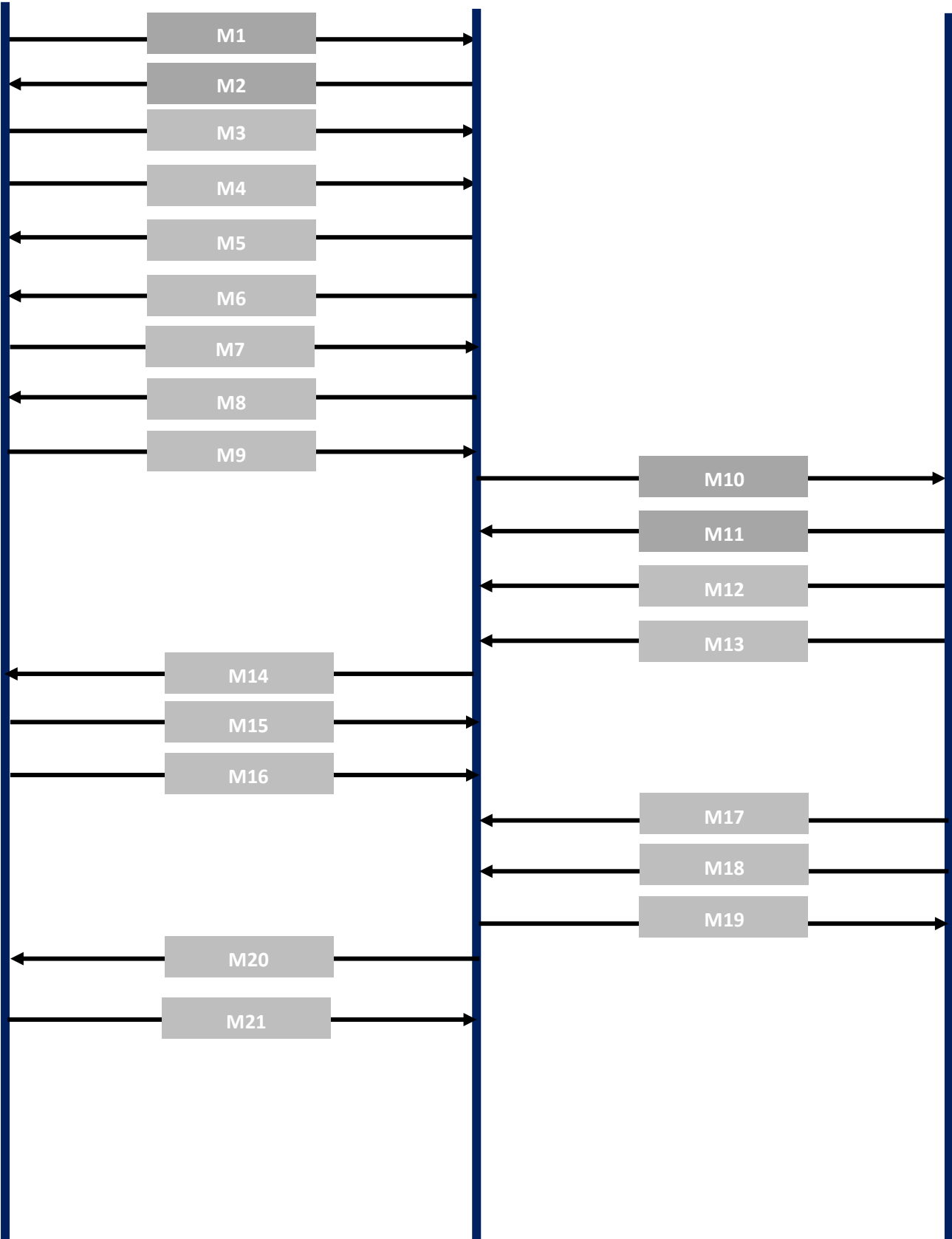
12			(RTCP) Receiver Report Source description	Page
13			(RTP) PT=ITU-T G.711 PCMU, SSRC=0xAD00A6E7, Seq=5819, Time=139100, Mark	Page
⋮				
14		(SIP) Status: 200 OK (INVITE)		Page
15	(SIP) Request: ACK sip:2222@178.14.3.13			Page
16	(UDP) 46595 → 5060			Page
⋮				
17			(UDP) 3269 → 5060 Len=4	Page
18			(SIP) Request: BYE sip:1111@178.14.3.13	Page
19		(SIP) Status: 200 OK (BYE)		Page
20		(SIP) Request: BYE sip:1111@178.14.3.14:46595		Page
21	(SIP) Status: 200 OK (BYE)			Page

Flow Chart

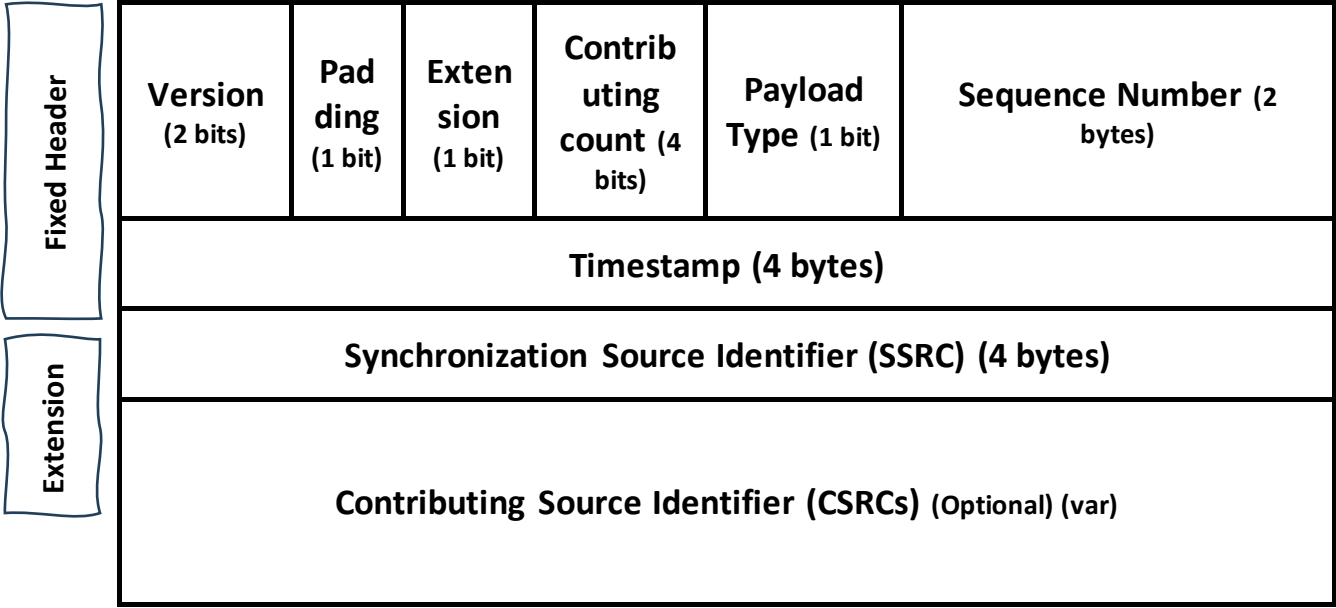
Client 1

Server

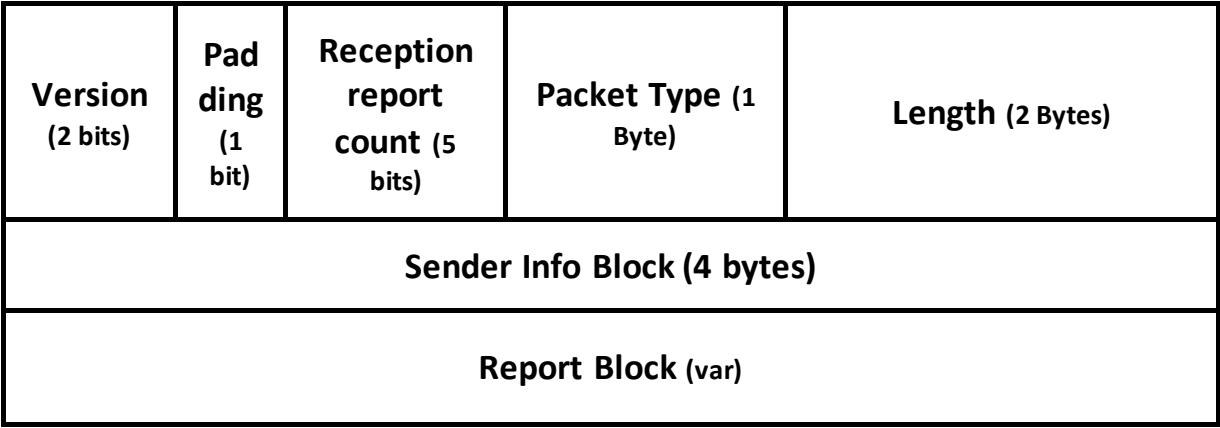
Client 2



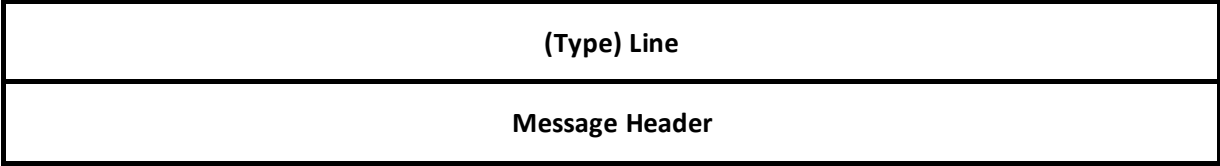
RTP Format



RTCP Format



SIP Format (Type)



M₁ (SIP/SDP Request)

0020 ... 49 4e 56 49 54 45
0030 20 73 69 70 3a 32 32 32 32 40 6c 6f 63 61 6c 68
0040 6f 73 74 2e 6c 6f 63 61 6c 64 6f 6d 61 69 6e 20
0050 53 49 50 2f 32 2e 30 0d 0a 56 69 61 3a 20 53 49
0060 50 2f 32 2e 30 2f 55 44 50 20 31 37 38 2e 31 34
0070 2e 33 2e 31 34 3a 34 36 35 39 35 3b 62 72 61 6e
0080 63 68 3d 7a 39 68 47 34 62 4b 2d 64 38 37 35 34
0090 33 2d 39 37 31 66 30 38 31 30 33 66 33 34 62 31
00a0 37 36 2d 31 2d 2d 64 38 37 35 34 33 2d 3b 72 70
00b0 6f 72 74 0d 0a 4d 61 78 2d 46 6f 72 77 61 72 64
00c0 73 3a 20 37 30 0d 0a 43 6f 6e 74 61 63 74 3a 20
00d0 3c 73 69 70 3a 31 31 31 31 40 31 37 38 2e 31 34
00e0 2e 33 2e 31 34 3a 34 36 35 39 35 3e 0d 0a 54 6f
00f0 3a 20 22 32 32 32 32 22 3c 73 69 70 3a 32 32 32
0100 32 40 6c 6f 63 61 6c 68 6f 73 74 2e 6c 6f 63 61
0110 6c 64 6f 6d 61 69 6e 3e 0d 0a 46 72 6f 6d 3a 20
0120 22 31 31 31 31 22 3c 73 69 70 3a 31 31 31 31 40
0130 6c 6f 63 61 6c 68 6f 73 74 2e 6c 6f 63 61 6c 64
0140 6f 6d 61 69 6e 3e 3b 74 61 67 3d 38 37 37 30 66
0150 66 32 38 0d 0a 43 61 6c 6c 2d 49 44 3a 20 59 6a
0160 45 34 4d 6d 55 30 59 7a 55 77 5a 54 5a 6c 4e 6d
0170 45 31 59 54 4a 68 4e 47 59 35 5a 6a 41 79 4e 6a
0180 5a 6c 4e 6d 4d 7a 5a 57 55 2e 0d 0a 43 53 65 71
0190 3a 20 31 20 49 4e 56 49 54 45 0d 0a 41 6c 6c 6f
01a0 77 3a 20 49 4e 56 49 54 45 2c 20 41 43 4b 2c 20
01b0 43 41 4e 43 45 4c 2c 20 4f 50 54 49 4f 4e 53 2c
01c0 20 42 59 45 2c 20 52 45 46 45 52 2c 20 4e 4f 54
01d0 49 46 59 2c 20 4d 45 53 53 41 47 45 2c 20 53 55
01e0 42 53 43 52 49 42 45 2c 20 49 4e 46 4f 0d 0a 43
01f0 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 61 70 70
0200 6c 69 63 61 74 69 6f 6e 2f 73 64 70 0d 0a 55 73
0210 65 72 2d 41 67 65 6e 74 3a 20 58 2d 4c 69 74 65
0220 20 72 65 6c 65 61 73 65 20 31 30 30 36 65 20 73
0230 74 61 6d 70 20 33 34 30 32 35 0d 0a 43 6f 6e 74
0240 65 6e 74 2d 4c 65 6e 67 74 68 3a 20 33 32 32 0d
0250 0a 0d 0a ...

INVITE sip:2222@localhost.localdomain SIP/2.0

Via: SIP/2.0/UDP 178.14.3.14:46595;branch=z9hG4bK-d87543-971f08103f34b176-1--d87543-
;rport

Max-Forwards: 70

Contact: <sip:1111@178.14.3.14:46595>

To: "2222"<sip:2222@localhost.localdomain>

From: "1111"<sip:1111@localhost.localdomain>;tag=8770ff28

Call-ID: YjE4MmU0YzUwZTZlNmE1YTJhNGY5ZjAyNjZlNmMzZWU.

[Generated Call-ID: YjE4MmU0YzUwZTZlNmE1YTJhNGY5ZjAyNjZlNmMzZWU.]

CSeq: 1 INVITE

Allow: INVITE, ACK, CANCEL, OPTIONS, BYE, REFER, NOTIFY, MESSAGE, SUBSCRIBE, INFO

Content-Type: application/sdp

User-Agent: X-Lite release 1006e stamp 34025

Content-Length: 322

M2 (SIP Status)

0000 ... 53 49 50 2f 32 2e 30 20 34 30 37 20 50 72 6f 78
0010 79 20 41 75 74 68 65 6e 74 69 63 61 74 69 6f 6e
0020 20 52 65 71 75 69 72 65 64 0d 0a 56 69 61 3a 20
0030 53 49 50 2f 32 2e 30 2f 55 44 50 20 31 37 38 2e
0040 31 34 2e 33 2e 31 34 3a 34 36 35 39 35 3b 62 72
0050 61 6e 63 68 3d 7a 39 68 47 34 62 4b 2d 64 38 37
0060 35 34 33 2d 39 37 31 66 30 38 31 30 33 66 33 34
0070 62 31 37 36 2d 31 2d 2d 64 38 37 35 34 33 2d 3b
0080 72 65 63 65 69 76 65 64 3d 31 37 38 2e 31 34 2e
0090 33 2e 31 34 3b 72 70 6f 72 74 3d 34 36 35 39 35
00a0 0d 0a 46 72 6f 6d 3a 20 22 31 31 31 22 3c 73
00b0 69 70 3a 31 31 31 31 40 6c 6f 63 61 6c 68 6f 73
00c0 74 2e 6c 6f 63 61 6c 64 6f 6d 61 69 6e 3e 3b 74
00d0 61 67 3d 38 37 37 30 66 66 32 38 0d 0a 54 6f 3a
00e0 20 22 32 32 32 32 22 3c 73 69 70 3a 32 32 32 32
00f0 40 6c 6f 63 61 6c 68 6f 73 74 2e 6c 6f 63 61 6c
0100 64 6f 6d 61 69 6e 3e 3b 74 61 67 3d 61 73 33 30
0110 61 32 62 66 62 38 0d 0a 43 61 6c 6c 2d 49 44 3a
0120 20 59 6a 45 34 4d 6d 55 30 59 7a 55 77 5a 54 5a
0130 6c 4e 6d 45 31 59 54 4a 68 4e 47 59 35 5a 6a 41
0140 79 4e 6a 5a 6c 4e 6d 4d 7a 5a 57 55 2e 0d 0a 43
0150 53 65 71 3a 20 31 20 49 4e 56 49 54 45 0d 0a 55
0160 73 65 72 2d 41 67 65 6e 74 3a 20 41 73 74 65 72
0170 69 73 6b 20 50 42 58 0d 0a 41 6c 6c 6f 77 3a 20
0180 49 4e 56 49 54 45 2c 20 41 43 4b 2c 20 43 41 4e
0190 43 45 4c 2c 20 4f 50 54 49 4f 4e 53 2c 20 42 59
01a0 45 2c 20 52 45 46 45 52 2c 20 53 55 42 53 43 52
01b0 49 42 45 2c 20 4e 4f 54 49 46 59 0d 0a 53 75 70
01c0 70 6f 72 74 65 64 3a 20 72 65 70 6c 61 63 65 73
01d0 0d 0a 50 72 6f 78 79 2d 41 75 74 68 65 6e 74 69
01e0 63 61 74 65 3a 20 44 69 67 65 73 74 20 61 6c 67
01f0 6f 72 69 74 68 6d 3d 4d 44 35 2c 20 72 65 61 6c
0200 6d 3d 22 61 73 74 65 72 69 73 6b 22 2c 20 6e 6f
0210 6e 63 65 3d 22 37 32 61 64 32 63 39 37 22 0d 0a
0220 43 6f 6e 74 65 6e 74 2d 4c 65 6e 67 74 68 3a 20
0230 30 0d 0a 0d 0a ...

Status-Line: SIP/2.0 407 Proxy Authentication Required

Via: SIP/2.0/UDP 178.14.3.14:46595;branch=z9hG4bK-d87543-971f08103f34b176-1--d87543-
;received=178.14.3.14;rport=46595

From: "1111"<sip:1111@localhost.localdomain>;tag=8770ff28

To: "2222"<sip:2222@localhost.localdomain>;tag=as30a2bfb8

Call-ID: YjE4MmU0YzUwZTZlNmE1YTJhNGY5ZjAyNjZlNmMzZWU.

[Generated Call-ID: YjE4MmU0YzUwZTZlNmE1YTJhNGY5ZjAyNjZlNmMzZWU.]

CSeq: 1 INVITE

User-Agent: Asterisk PBX

Allow: INVITE, ACK, CANCEL, OPTIONS, BYE, REFER, SUBSCRIBE, NOTIFY

Supported: replaces

Proxy-Authenticate: Digest algorithm=MD5, realm="asterisk", nonce="72ad2c97"

Content-Length: 0

M₈ (RTP Stream Setup)

0000 ... 80 00 39 43 00 02 1f 58 7d 42 aa 5e ...

RFC 1889 Version (2)	False	False	0	Payload type: ITU-T G.711 PCMU (0)	14659
139096					
2101520990					

M₉ (RTCP Receiver Report)

0020 ... 80 c9 00 01 d8 53

0030 41 9c 81 ca 00 1e d8 53 41 9c 01 3d 31 34 36 33
0040 42 36 43 44 42 43 43 42 34 36 44 42 38 42 36 41
0050 38 31 34 46 35 34 39 43 32 32 34 38 40 75 6e 69
0060 71 75 65 2e 7a 41 34 41 34 35 32 41 44 45 41 34
0070 31 34 34 31 42 2e 6f 72 67 08 31 10 78 2d 72 74
0080 70 2d 73 65 73 73 69 6f 6e 2d 69 64 35 36 38 34
0090 42 46 39 32 35 38 38 43 34 39 31 39 41 44 30 38
00a0 41 43 35 38 41 43 44 44 35 41 33 33 00 00

RFC 1889 Version (2)	False	0	Receiver Report (201)	1
3629334940				
Real-time Transport Control Protocol (Source description) [Stream setup by SDP (frame 10)] 10.. = Version: RFC 1889 Version (2) ..0. = Padding: False ...0 0001 = Source count: 1 Packet type: Source description (202) Length: 30 (124 bytes) Chunk 1, SSRC/CSRC 0xD853419C [RTCP frame length check: OK - 132 bytes]				